Chapter 6. Permutation Tests for Analysis of Variance

Kruskal–Wallis, ordered alternatives, Friedman test, general two-way layouts

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Overview

This chapter covers permutation tests in the analysis of variance (one-way and two-way layouts). Recall that distribution-free tests for comparing two populations were permutation tests after rank transformation; the distribution-free tests introduced here are also permutation tests after rank transformation. We introduce the Kruskal–Wallis test and Jonckheere test for the one-way layout, and the Friedman test, Page test and Skillings–Mack test for the two-way layout.[1]

6.1 One-way layout: comparing several populations

Consider comparing several populations or the effects of several treatments.

Example 6.1

The usage times of shared electric kickboards in service around Jamsil, Seoul, were surveyed among citizens who used one between 1 and 3 pm on a weekend.

company (brand)	usage time (min)	number of users
Kickgoing	10, 10, . . . , 50, 51	32
Lime	10, 10, 10, . . . , 103, 108	101
Gogossing	10, 10, . . . , 33, 44	22

Usage times under 10 minutes were recorded as 10 minutes. The data are `kickboard.csv`; their distributions are compared by box plots in the book. Do the usage times of the three companies' kickboards differ by service company?

Example 6.1 compares three populations of riders of three different brands of electric kickboards: a kind of population model as in Chapter 3.

Example 6.2

To shorten the time needed to bake a pizza, a student designed an experiment on the time sugar and milk take to activate the yeast. Four recipes (A, B, C, D) were assigned at random to 16 batches of dough, and the time until the dough rose was measured (`pizza.csv`).

recipe	time (s)
A	575, 395, 525, 450
B	280, 120, 175, 210
C	720, 554, 700, 650
D	200, 135, 250, 150

Do the four recipes differ? If so, which recipe activates fastest?

Example 6.2 compares the effects of four different treatments: a kind of randomization model. In experimental design such an experiment is called a **one-way layout** (one-way factorial design). There is one *factor* (here the recipe) affecting the result, and its values are one of A, B, C, D. Since one factor is laid out, it is called a one-way layout. In the population model of Example 6.1 each population can be seen as a value of the factor "company (brand)," so it is the same model as a one-way layout. In statistics both situations are called one-way layouts.

[1] Implementing this chapter in R needs various statistics and shuffles. For simplicity of the examples several R functions are collected in `np_supp.R`; run `source("np_supp.R")` in R to use them.

The data of a one-way layout can be written as follows. Suppose the factor takes k distinct values, the *(factor) levels*; e.g. the factor "recipe" of Example 6.2 has $k = 4$ levels A, B, C, D. If there are n_1 observations $X_{11}, \ldots, X_{1n_1}$ at the first level, n_2 observations $X_{21}, \ldots, X_{2n_2}$ at the second, and so on, the data and their descriptive statistics are:

level	observations	sample size	sample mean	sample variance
1	$X_{11}, \ldots, X_{1n_1}$	n_1	$\bar{X}_1$	S_1^2
2	$X_{21}, \ldots, X_{2n_2}$	n_2	$\bar{X}_2$	S_2^2
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
k	$X_{k1}, \ldots, X_{kn_k}$	n_k	$\bar{X}_k$	S_k^2

The total number of observations is $n = \sum_{i=1}^{k} n_i$. Assuming the observations are independent, model

$$X_{ij} = \mu_i + \epsilon_{ij}, \qquad i = 1, \ldots, k, \quad j = 1, \ldots, n_i. \tag{6.1}$$

The jth observation of level i has location μ_i and error ϵ_{ij}; all errors follow a distribution with mean 0 (or median 0). We test whether the location parameters μ_i are equal or different:

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k \quad \text{vs} \quad H_1 : \mu_i \neq \mu_j \text{ for some } i, j.$$

The traditional one-way layout assumes normal errors with mean 0, but sometimes this normality is severely violated. E.g. the kickboard usage times of Example 6.1 have a long right tail with most users finishing within 10–20 minutes, a very skewed distribution, so normality cannot be assumed, and the CLT approximation applies only for very large samples.

We first look at the traditional method under normality, the analysis of variance (ANOVA), then the nonparametric (distribution-free) permutation test, and the rank-based permutation test, the Kruskal–Wallis test.

6.1.1 Classical analysis of variance

In classical ANOVA the different levels of the factor are called different **treatments** (e.g. recipes). ANOVA compares the variability of observations due to different factor levels (i.e. due to different treatments) with the variability due to error, using an F-statistic. To describe it simply, assume normality $X_{ij} \sim N(\mu_i, \sigma^2)$.

The variability due to different treatments is measured by the variability of the treatment means $\{\bar{X}_1, \ldots, \bar{X}_k\}$. Comparing them with the grand mean

$$\bar{X} = \frac{1}{n} \sum_{i=1}^{k} \sum_{j=1}^{n_i} X_{ij} = \frac{1}{n} \sum_{i=1}^{k} n_i \bar{X}_i,$$

the treatment variability is

$$SS_{\text{Treatment}} = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (\bar{X}_i - \bar{X})^2 = \sum_{i=1}^{k} n_i (\bar{X}_i - \bar{X})^2. \tag{6.2}$$

Since $E(\bar{X}_i) = \mu_i$, $SS_{\text{Treatment}}$ tends to be large when the μ_i differ a lot. To standardize its size it is compared with the error variability

$$SS_{\text{Error}} = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_i)^2.$$

The ratio of the mean variabilities (each divided by its degrees of freedom) is the F-statistic

$$F = \frac{SS_{\text{Treatment}}/(k-1)}{SS_{\text{Error}}/(n-k)}. \tag{6.3}$$

When the alternative is true (μ_i differ), F tends to be large.

If $H_0 : \mu_1 = \cdots = \mu_k$ is true and the populations are normal (with equal variances), all observations X_{ij} are independent with the same normal distribution, and F is known to follow the F-distribution $F_{k-1,n-k}$ with degrees of freedom $k-1, n-k$. (I.e. the null distribution of F is $F_{k-1,n-k}$.) With observed F_{obs}, the ANOVA p-value is the probability of an F-statistic at least F_{obs}, $p = P(F_{k-1,n-k} \geq F_{\text{obs}})$. This test is the ANOVA F-test, or simply analysis of variance.

Why $F \sim F_{k-1,n-k}$ (supplied). Under H_0 all $X_{ij} \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$. Within level i, $\sum_j (X_{ij} - \bar{X}_i)^2/\sigma^2 \sim \chi^2_{n_i-1}$, independent of $\bar{X}_i$; the levels are independent, so $SS_{\text{Error}}/\sigma^2 \sim \chi^2_{n-k}$, independent of $(\bar{X}_1, \ldots, \bar{X}_k)$. The vector $(\sqrt{n_i}(\bar{X}_i - \mu)/\sigma)_i$ is standard normal in $\mathbb{R}^k$, and $SS_{\text{Treatment}}/\sigma^2$ is its squared length after projection onto the orthogonal complement of $(\sqrt{n_1}, \ldots, \sqrt{n_k})$, so it is χ^2_{k-1} (Cochran). The ratio of two independent χ^2's divided by their degrees of freedom is $F_{k-1,n-k}$. ■

Example 6.3

With the kickboard usage data of Example 6.1, let μ_i be the mean usage time of each of the three brands and run the F-test of $H_0 : \mu_1 = \mu_2 = \mu_3$. The $n = 155$ observations come at $k = 3$ factor levels; the treatment variability is $SS_{\text{Treatment}} = 739$, the error variability $SS_{\text{Error}} = 30287$, and the F-statistic 1.855. Compared with $F_{2,152}$ the p-value is 0.16: not enough evidence that mean usage time differs by brand. In R with `aov()`:

```
> kick = read.csv("data/kickboard.csv", header = TRUE)
> aov(time ~ brand, data = kick) %>% summary()
             Df Sum Sq Mean Sq F value Pr(>F)
brand         2    739   369.7   1.855   0.16
Residuals   152  30287   199.3
```

For reference, the total variability $SS_{\text{Total}} = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (X_{ij} - \bar{X})^2$ is the sum of treatment and error variability:

$$SS_{\text{Total}} = SS_{\text{Treatment}} + SS_{\text{Error}}. \tag{6.4}$$

Proof (supplied). Write $X_{ij} - \bar{X} = (X_{ij} - \bar{X}_i) + (\bar{X}_i - \bar{X})$, square and sum. The cross term is $2\sum_i (\bar{X}_i - \bar{X}) \sum_j (X_{ij} - \bar{X}_i) = 0$, since $\sum_j (X_{ij} - \bar{X}_i) = 0$ for each i. ■

Dividing each variability by its degrees of freedom gives sample variances, hence the name analysis of variance.

Permutation tests in ANOVA. When the populations resemble normal distributions, the ANOVA F-test controls the type I error well at the given level and has high power. But when the populations are very skewed, as in Example 6.1, normality is badly violated and the F-test cannot be said to control the type I error well. Then one can use the permutation test with the F-statistic (6.3). Recall that a permutation test, whatever statistic it uses, is exact (controls the type I error at the given level) as long as the exchangeability under the null holds.

In the one-way model (6.1), leave the distribution of ϵ_{ij} unspecified and assume only that it is a distribution with median 0; for Example 6.1 a right-skewed distribution seems the distribution of ϵ_{ij}. When $H_0 : \mu_1 = \cdots = \mu_k$ is true, the observations $\{X_{11}, \ldots, X_{kn_k}\}$ are independent with the same distribution, so the exchangeability needed for the permutation test holds. To consider all possible shuffles under the null, it is convenient to present the data as pairs (observation, factor level):

```
> pizza <- read.delim("data/pizza.txt")
> pizza
   Activation.Times Recipe
1               575      A
2               395      A
3               280      B
4               120      B
5               720      C
6               554      C
7               200      D
8               135      D
9               525      A
10              450      A
11              175      B
12              210      B
13              700      C
14              650      C
15              250      D
16              150      D
```

These are the data of Example 6.2, the numerical observations (X_{ij}; `Activation.Times`) split by the categorical factor ($i = A, B, C, D$; `Recipe`). When the null is true, the factor level of each observation is just the result of one random way of splitting the 16 observations into 4 subgroups of size 4, and any shuffle of `Recipe` would have arisen with the same probability under the null.

In other words, when the null is true and the data values X_{ij} are fixed, the classification of the observations by factor levels $1, \ldots, k$ is one of the

$$\binom{n}{n_1, \ldots, n_k} = \frac{n!}{n_1! \cdots n_k!}$$

ways of splitting the n observations into subgroups of sizes $n_1, \ldots, n_k$. So the null distribution of a statistic measuring how extreme the observed data are under the null can be generated from all $\binom{n}{n_1,\ldots,n_k}$ shuffles. In Example 6.2 there are only $n = 16$ observations, but the possible shuffles number about 63 million[2], quite many, so a random permutation test is recommended.

Example 6.4

For the pizza data of Example 6.2, approximating the permutation distribution (null distribution) of F with $M = 9999$ random shuffles and comparing with $F_{\text{obs}} = 44.7$ gives Figure 6.1 of the book.

```
> (Fobs = F.stat(Activation.Times, Recipe, data = pizza))
[1] 44.73917
> set.seed(0)
> M = 9999
> Fperm = replicate(M, {
+   perm.x = sample(pizza$Activation.Times, replace = FALSE)
+   F.stat(perm.x, pizza$Recipe)
+ })
```

(The function `F.stat()` is in `np_supp.R`.) In Figure 6.1 the histogram of the F-statistics f_i from 9999 random shuffles is the (approximate) permutation distribution of F. F_{obs} is very large, so the permutation p-value is

$$p = \frac{1 + \sum_{i=1}^{M} I(f_i \geq F_{\text{obs}})}{1 + M} = \frac{1 + 0}{1 + M} = 0.0001.$$

[2] $\binom{16}{4,4,4,4} = 63063000$.

```
> (p.val = mean(c(Fobs,Fperm) >= Fobs))
[1] 1e-04
```

Zooming in on $F \in [0, 5]$ in Figure 6.1 shows, in blue, the theoretical distribution $F_{3,12}$ of the statistic under normality; the permutation distribution and $F_{3,12}$ are very similar.

Since this permutation test uses the F-statistic, it is also called the permutation F-test.[3] When the populations are close to normal, the permutation F-test and the F-test (assuming normality) give roughly similar results. When normality is badly violated, the permutation test controls the type I error but the F-test is not exact.

6.1.2 The Kruskal–Wallis test

Permutation tests can use statistics other than F. We look at a rank-based permutation test.

Recall how the observations were rank-transformed in the Wilcoxon rank-sum test, the rank-based test for comparing two populations. When the null "the two populations are the same" is true, all observations were sampled at random, so the observations of the pooled sample were ranked from smallest to largest. (With ties of two or more observations, mid-ranks were given.)

For comparing two or more populations likewise, arrange all observations $X_{11}, \ldots, X_{kn_k}$ in order and rank them from the smallest, ignoring which population each came from. E.g. the ranks of the pizza data of Example 6.2:

```
> (pizza = pizza %>% mutate(rank = rank(Activation.Times)))
   Activation.Times Recipe rank
1               575      A   13
2               395      A    9
3               280      B    8
4               120      B    1
5               720      C   16
6               554      C   12
7               200      D    5
8               135      D    2
9               525      A   11
10              450      A   10
11              175      B    4
12              210      B    6
13              700      C   15
14              650      C   14
15              250      D    7
16              150      D    3
```

Kruskal and Wallis (Kruskal and Wallis, 1952) proposed the permutation test using as statistic the F-statistic computed from the rank-transformed values $X_{ij} \mapsto R_{ij}$, the **Kruskal–Wallis (one-way) test**. Concretely, with R_{ij} the rank of X_{ij}, the rank F-statistic is

$$F_R = \frac{\sum_{i=1}^{k} n_i(\bar{R}_i - \bar{R})^2/(k-1)}{\sum_{i=1}^{k}\sum_{j=1}^{n_i}(R_{ij} - \bar{R}_i)^2/(n-k)}, \tag{6.5}$$

the F-statistic computed as if the ranks $R_{11}, \ldots, R_{kn_k}$ were the observations. Kruskal and Wallis actually used the Kruskal–Wallis statistic (KW statistic), which gives the same test result as F_R:

$$KW = \frac{12}{n(n+1)} \sum_{i=1}^{k} n_i \Big(\bar{R}_i - \frac{n+1}{2}\Big)^2. \tag{6.6}$$

[3] Using $SS_{\text{Treatment}}$ or SS_{Error} instead of F gives the same permutation test, because by (6.4) SS_{Total} is the same for all shuffles: $F = \frac{n-k}{k-1} \frac{SS_{\text{Treatment}}}{SS_{\text{Total}} - SS_{\text{Treatment}}}$ is increasing in $SS_{\text{Treatment}}$.

Recalling that the ranks $\{1,\dots,n\}$ have mean $\bar{R} = \frac{1}{n}\sum_{i=1}^{n} i = \frac{n+1}{2}$ and sample variance $S_R^2 = \frac{1}{n-1}\sum_{i=1}^{n}(i-\bar{R})^2 = \frac{n(n+1)}{12}$, the KW statistic is the treatment variability of the ranks $\sum_{i=1}^{k} n_i(\bar{R}_i - \frac{n+1}{2})^2$ divided by the sample variance S_R^2 of the ranks. With tied observations, the sample variance of the ranks is not exactly $n(n+1)/12$, so to cover ties too the KW statistic is defined as

$$KW = \frac{\sum_{i=1}^{k} n_i\left(\bar{R}_i - \frac{n+1}{2}\right)^2}{S_R^2}, \tag{6.7}$$

where S_R^2 in (6.7) is the sample variance of the tie-corrected ranks $\{R_{11},\dots,R_{kn_k}\}$.

F_R and KW give the same test (supplied). For the ranks, $SS^R_{\text{Total}} = \sum_{i,j}(R_{ij}-\bar{R})^2 = (n-1)S_R^2$ is the same for every shuffle, and $SS^R_{\text{Treatment}} = \sum_i n_i(\bar{R}_i - \bar{R})^2 = S_R^2 \cdot KW$. By (6.4) applied to the ranks,

$$F_R = \frac{n-k}{k-1}\cdot\frac{SS^R_{\text{Treatment}}}{SS^R_{\text{Total}} - SS^R_{\text{Treatment}}} = \frac{n-k}{k-1}\cdot\frac{KW}{n-1-KW},$$

a strictly increasing function of KW on $[0, n-1)$. So the two statistics order the shuffles identically and give the same permutation p-value. ■

When the null is true there is no location difference between the populations, so the ranks R_{ij} take (without ties) one of $\{1,\dots,n\}$ at random. So the treatment mean ranks $\bar{R}_i$ are expected near the overall mean $\bar{R} = \frac{n+1}{2}$ and the KW statistic is expected to be small. When the alternative is true, e.g. if the third population lies to the right of the others, its mean rank $\bar{R}_i$ is expected to be large too: under the alternative the KW statistic tends to be large.

When the null is true, the null distribution of KW is obtained from its permutation distribution. The histogram in Figure 6.2 of the book approximates the null distribution of KW for the situation of Example 6.2, $n = 16$, $n_1 = n_2 = n_3 = n_4 = 4$, by $M = 9999$ random shuffles. (The overlaid density is its large-sample approximation, explained shortly.)

This permutation distribution has nothing to do with the exact values of the data X_{ij}, only with their relative positions. So it does not depend on the population distribution from which the data were sampled, and the test with the KW statistic is distribution-free. With observed KW statistic KW_{obs} the p-value is $p = P(K \geq KW_{\text{obs}})$ (with K following the permutation distribution of KW); with a random permutation test and shuffled values $k_1,\dots,k_M$ of KW,

$$p = \frac{1 + \sum_{i=1}^{M} I(k_i \geq KW_{\text{obs}})}{M+1}. \tag{6.8}$$

Example 6.5

For the pizza data of Example 6.2 the Kruskal–Wallis statistic is $KW_{\text{obs}} = 12.397$. Compared with the null distribution of KW (Figure 6.2), KW_{obs} is extreme, so the differences between recipes are statistically significant. The KW statistic can be computed by `KW.stat()` in `np_supp.R` or by R's built-in `kruskal.test()`:

```
> KW.stat(Activation.Times, Recipe, data = pizza)
[1] 12.39706
> kruskal.test(pizza$Activation.Times, pizza$Recipe)$statistic
Kruskal-Wallis chi-squared
                  12.39706
```

This statistic is called the Kruskal–Wallis "chi-squared" statistic; the next part explains why.

The Kruskal–Wallis chi-squared test. With several populations ($k \geq 3$), even a sample as small as $n = 16$ gives very many shuffles. Today, when computers make generating and approximating permutation distributions easy, using the (random) permutation distribution as the null

distribution is the right method; but in the 1950s, when the Kruskal–Wallis test was first developed, it was practically impossible. So Kruskal and Wallis devised approximating this permutation distribution for large n by the chi-squared distribution with $k-1$ degrees of freedom (χ^2_{k-1}). Using that under the null

$$KW \dot{\sim} \chi^2_{k-1},$$

the large-sample p-value is $p = P(\chi^2_{k-1} \geq KW_{\text{obs}})$.

To understand this approximation briefly: the numerator of the KW statistic (6.7) is the treatment variability with ranks as observations, $SS^R_{\text{Treatment}} := \sum_{i=1}^k n_i(\bar{R}_i - \bar{R})^2$. If the observations $X_{ij} \sim N(\mu, \sigma^2)$ were normal, each treatment mean $\bar{X}_i$ would also be normal $N(\mu, \sigma^2/n_i)$, and the treatment variability $SS_{\text{Treatment}} = \sum_{i=1}^k n_i(\bar{X}_i - \bar{X})^2$ is known to be exactly $\sigma^2\chi^2_{k-1}$.

Under the null, the ranks R_{ij} are not normal, but they are sampled uniformly from the finite population $\mathcal{A}_n := \{1, \ldots, n\}$, so by the (finite-population) CLT the $\bar{R}_i$ are asymptotically normal. So when the n_i are large, by the CLT, $SS^R_{\text{Treatment}}$ divided by the population variance σ^2_R of $\mathcal{A}_n$ is approximately χ^2_{k-1}; replacing σ^2_R by the sample variance S^2_R gives the KW statistic (6.7).

Sketch of $KW \xrightarrow{d} \chi^2_{k-1}$ (supplied). Let $Z_i = \sqrt{n_i}(\bar{R}_i - \bar{R})/S_R$. By Theorem 4.1 applied jointly (with the covariances of sampling without replacement), $\text{Var}(\bar{R}_i) = \frac{\sigma^2_R}{n_i}\frac{n-n_i}{n-1}$ and $\text{Cov}(\bar{R}_i, \bar{R}_l) = -\frac{\sigma^2_R}{n-1}$ for $i \neq l$, with $\sigma^2_R = \frac{n-1}{n}S^2_R$. Hence the covariance matrix of $(Z_1, \ldots, Z_k)$ tends to $I - \mathbf{u}\mathbf{u}'$ with $\mathbf{u} = (\sqrt{n_1/n}, \ldots, \sqrt{n_k/n})'$, a projection of rank $k-1$. By the multivariate finite-population CLT (Theorem 4.2 with the Cramér–Wold device), (Z_i) is asymptotically $N(0, I - \mathbf{u}\mathbf{u}')$, and $KW = \sum_i Z_i^2$ converges to the squared length of a standard normal vector projected onto a $(k-1)$-dimensional subspace, i.e. χ^2_{k-1}. ■

The blue curve in Figure 6.2 of the book ($n = 16$, $n_1 = \cdots = n_4 = 4$) is the density of the large-sample approximation χ^2_{4-1} of the permutation distribution of KW. A quantile plot comparing the permutation distribution with χ^2_3 (Figure 6.3 of the book) shows that the large-sample approximation is very inaccurate, especially in the tails. Note that for small samples the Kruskal–Wallis χ^2 test is not exact.

Figure 6.4 of the book compares the permutation distribution of KW with χ^2_2 when $(n_1, n_2, n_3) = (32, 101, 22)$ (exactly the sample sizes of Example 6.1). With many observations, the null distribution of KW (histogram) is well approximated by χ^2_2.

Example 6.6

For the data of Example 6.1 the KW statistic is $KW_{\text{obs}} = 2.67$, and the p-value of the Kruskal–Wallis chi-squared test against the large-sample distribution χ^2_2 is $p = P(\chi^2_2 \geq 2.67) = 0.2634$. This p-value is given directly by `kruskal.test()`:

```
> (KWobs = KW.stat(time, brand, data = kick))
[1] 2.668463
> (pvalue.chisq = 1 - pchisq(KWobs, df = 2))
[1] 0.2633605
> kruskal.test(time ~ brand, kick)

        Kruskal-Wallis rank sum test

data:  time by brand
Kruskal-Wallis chi-squared = 2.6685, df = 2, p-value = 0.2634
```

With many observations, the random permutation p-value and the large-sample p-value should be similar; the random permutation p-value $p = 0.2765$ leads to almost the same conclusion.

```
> set.seed(0)
> M = 9999
```

```
> KWperm = replicate(M, {
+   x.perm = sample(kick$time, replace = FALSE)
+   KW.stat(x.perm, kick$brand)
+ })
> (p.val = mean(c(KWobs,KWperm) >= KWobs))
[1] 0.2765
```

6.1.3 Ordered alternatives

When comparing several populations, the populations may follow some order, and we may hypothesize that the locations of observations increase or decrease accordingly.

Example 6.7

For 16 oak trees on Gwanak Mountain the annual increase of trunk diameter was measured, and each tree was classified by the slope of the place where it stands (`tree.csv`).

slope	annual increase of trunk diameter (cm)
slope I (steepest)	1.91, 1.53, 2.08, 1.71
slope II	2.44
slope III	2.45, 2.04, 1.60, 2.37
slope IV	2.52, 2.36, 2.73
slope V (no slope)	2.78, 2.88, 2.10, 1.66

For the data of Example 6.7, comparing five populations, we hypothesize that the steeper the slope, the smaller the growth of the oak. Regard the five slopes as five treatments and let $\mu_1, \ldots, \mu_5$ be the mean treatment effects. In the one-way model (6.1), against the null

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5,$$

the alternative can be set as

$$H_1 : \mu_1 < \mu_2 < \mu_3 < \mu_4 < \mu_5.$$

[4] When the alternative is this specific, a test with higher power than the F-test or the Kruskal–Wallis test, which consider broad alternatives, can be devised.

When H_1 is true, for any pair of populations $i < j$ implies $\mu_i < \mu_j$, so define a statistic T_{ij} that allows a one-sided test of the locations of the two populations. E.g. with $T_{ij} := \bar{X}_j - \bar{X}_i$, T_{ij} is expected to be large when $\mu_j > \mu_i$. When the alternative is true, all T_{ij} with $i < j$ are large, so

$$T = \sum_{i<j} T_{ij} \tag{6.9}$$

also tends to be large. When the null is true, the T_{ij} are around 0.

To make the statistic (6.9) distribution-free, Jonckheere (1954) proposed taking as pairwise statistic T_{ij} the Mann–Whitney U-statistic

$$T_{ij} = \sum_{\ell_i=1}^{n_i} \sum_{\ell_j=1}^{n_j} I(X_{i\ell_i} < X_{j\ell_j}).$$

[4] More generally, with F_i the distribution function of the ith population, one may set $H_0 : F_1(x) = F_2(x) = \cdots = F_k(x)$ and $H_1 : F_1(x) \geq F_2(x) \geq \cdots \geq F_k(x)$ (at least one inequality strict); then the kth population tends to have the largest values. Or, if the first population tends to have the largest values, $H_1 : F_1(x) \leq \cdots \leq F_k(x)$.

The permutation test with the statistic T (6.9) with Mann–Whitney U-statistics as T_{ij} is the **Jonckheere test** or **Jonckheere–Terpstra test**. In the Jonckheere–Terpstra test, using Wilcoxon rank-sum statistics instead of Mann–Whitney U-statistics gives the same result. Terpstra (1952) showed that the permutation test with T_{ij} is the same as the permutation test using Kendall's tau statistic τ_n, which measures nonparametrically the correlation between the observations X_{ij} and the population order $\{1, \ldots, k\}$.

The null distribution of the Jonckheere statistic can be obtained by random permutations; for large samples a large-sample approximation is also used. This approximation is the same as the large-sample approximation of Kendall's tau statistic, so the discussion is omitted here.

Example 6.8

In the hypothesis test about the oaks of Example 6.7, the Kruskal–Wallis chi-squared test of the general alternative H_1: "$\mu_i \neq \mu_j$ for some $i \neq j$" ("the growth rate of the trees differs by slope") has p-value 0.2016 and does not reject $H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$. Setting the alternative more specifically as the ordered hypothesis "the lower the slope, the better the trees grow" ($H_1 : \mu_1 < \mu_2 < \mu_3 < \mu_4 < \mu_5$), the Jonckheere test has p-value 0.015 and rejects the null at the 5% level.

To run the Jonckheere test in R use `JonckheereTerpstraTest()` of the `DescTools` package. To use this function, the variable for the treatment must first be an "ordered" factor.

```
> tree = read.csv("data/tree.csv")
> head(tree)
  increment slope
1      1.91     1
2      1.53     1
3      2.08     1
4      1.71     1
5      2.44     2
6      2.45     3
> # this is needed for JT test
> (tree$slope = factor(tree$slope, ordered = TRUE))
 [1] 1 1 1 1 2 3 3 3 3 4 4 4 5 5 5 5
Levels: 1 < 2 < 3 < 4 < 5
```

The Jonckheere test implemented in `DescTools::JonckheereTerpstraTest()` gives the result of the exact permutation test generating all shuffles for $n \leq 100$, and the large-sample approximate p-value for $n > 100$.

```
> with(tree, JonckheereTerpstraTest(increment , slope, alternative =
    "increasing"))

        Jonckheere-Terpstra test

data:  increment and slope
JT = 73, p-value = 0.01534
alternative hypothesis: increasing
```

The oak data of Examples 6.7–6.8 have $n = 16$, so this p-value is that of the exact permutation test. To run a random permutation test, specify the argument `nperm`:

```
with(tree, JonckheereTerpstraTest(increment , slope, alternative = "
    increasing", nperm= 5000))
```

6.2 Inference in complete block designs without replication

So far we have compared several populations or several treatments. If, besides the treatment, another variable affects the observed effect, *blocking* is used to control it. Blocking compares treatments within homogeneous subgroups when the population to which the treatments are applied is not homogeneous. E.g. to compare the efficacy of a newly developed vaccine with a placebo, the effect may differ between healthy adults and adolescents; the treatments are vaccine/placebo and the block variable splits the whole population into the two subgroups adults/adolescents.

Example 6.9

Three diets A, B, C are tested on animals in four nutritional states. Three animals are obtained for each nutritional state, the diets are applied as in the design below, and the weight gains are recorded.

state I	state II	state III	state IV
A	C	B	A
B	A	C	C
C	B	A	B

A design like Example 6.9 is a (non-replicated) **complete block design** (CBD). Since the diets were assigned at random within each block (nutritional state), it is also called a randomized CBD. The measured weight gain mixes the gain due to the diet with the natural gain due to the nutritional state.

Example 6.10

An urban smart-farm entrepreneur growing lettuce wants to compare three cultivation methods (A, B, C). Growth may differ with the location of the smart farm, so with location (rooftop, inside a building, inside a subway station) as block variable, a randomized complete block design gave the observations below: production over 30 days (`smartfarm1.csv`).

	rooftop	inside building	inside subway station
A	85	105	95
B	95	96	110
C	100	112	117

Does production differ by cultivation method and by location of the smart farm?

In the two examples above, exactly one observation is obtained for each combination of treatment and block. Modeling the observations: with k treatments and b blocks, the observation X_{ij} of treatment i in block j is modeled as

$$X_{ij} = \mu + t_i + b_j + \epsilon_{ij}, \qquad i = 1, \ldots, k, \quad j = 1, \ldots, b. \tag{6.10}$$

Here μ is the effect unaffected by treatment or block, t_i the effect of the ith treatment, b_j the effect of the jth block, and ϵ_{ij} independent errors following a distribution with mean (or median) 0.

If several observations can be obtained for each treatment-block combination, it is a replicated complete block design, and can be seen as a somewhat more general two-way layout.

Example 6.11

For the smart-farm production of Example 6.10, several smart farms were installed at each location and more observations obtained. Does production differ by cultivation method and by location (`smartfarm.csv`)?

	rooftop	inside building	inside subway station
A	85, 88, 72	105, 100, 97	95, 69, 45
B	95, 100, 84	96, 87, 92	110, 94, 93
C	100, 105, 98	112, 109, 116	117, 121, 101

When repeated observations are possible, as in Example 6.11, not only the effects of treatments and blocks but also the treatment-block interaction can be modeled and estimated. The lth observation X_{ijl} of treatment i in block j is modeled as

$$X_{ijl} = \mu + t_i + b_j + (tb)_{ij} + \epsilon_{ijl}, \tag{6.11}$$

where $(tb)_{ij}$ is the interaction. The interaction could be modeled in (6.10) too, but with too few observations it cannot be estimated, so in complete block designs no interaction is assumed.

First consider tests of treatment effects in a complete block design with one observation per treatment and block, as in Examples 6.9 and 6.10. Assuming model (6.10), we compare the null

$$H_0 : t_1 = \cdots = t_k \tag{6.12}$$

(all treatment effects equal) with the alternative H_1: "not all treatment effects are equal." When the second variable is not a block but another kind of treatment, statistical inference proceeds the same way.

6.2.1 Classical analysis of variance

Writing the estimator of the effect of treatment i as $\bar{X}_{i\cdot} = \frac{1}{b}\sum_{j=1}^{b} X_{ij}$, of block j as $\bar{X}_{\cdot j} = \frac{1}{k}\sum_{i=1}^{k} X_{ij}$, and the grand mean as

$$\bar{X} = \bar{X}_{\cdot\cdot} = \frac{1}{kb}\sum_{i=1}^{k}\sum_{j=1}^{b} X_{ij} = \frac{1}{b}\sum_{j=1}^{b} \bar{X}_{\cdot j},$$

classical ANOVA writes the treatment variability as

$$SS_{\text{Treatment}} = \sum_{i=1}^{k}\sum_{j=1}^{b}(\bar{X}_{i\cdot} - \bar{X})^2 = b\sum_{i=1}^{k}(\bar{X}_{i\cdot} - \bar{X})^2.$$

In computing $\bar{X}_{i\cdot}$ and $\bar{X}$ the differences due to block effects cancel, so $SS_{\text{Treatment}}$ can be seen as treatment variability regardless of the block effects. Under the null (6.12), $SS_{\text{Treatment}}$ is expected to be small; when the alternative $t_i \neq t_{i'}$ is true it tends to be large. The F-statistic compares the treatment variability with the error variability SS_{Error}. In a complete block design the error is estimated after removing both treatment and block effects:

$$SS_{\text{Error}} = \sum_{i=1}^{k}\sum_{j=1}^{b}(X_{ij} - \bar{X}_{i\cdot} - \bar{X}_{\cdot j} + \bar{X})^2.$$

(The treatment effect $\bar{X}_{i\cdot}$ and block effect $\bar{X}_{\cdot j}$ both contain the grand mean $\bar{X}$, so to remove it only once it is added back.) The F-statistic is the ratio of the mean variabilities divided by their degrees of freedom:

$$F = \frac{SS_{\text{Treatment}}/(k-1)}{SS_{\text{Error}}/[(k-1)(b-1)]}. \tag{6.13}$$

When $H_0 : t_1 = \cdots = t_k$ is true and $\epsilon_{ij} \sim N(0, \sigma^2)$, F is known to follow the F-distribution with degrees of freedom $k-1$, $(k-1)(b-1)$, $F_{k-1,(k-1)(b-1)}$; so the p-value of F_{obs} is $p = P(F_{k-1,(k-1)(b-1)} \geq F_{\text{obs}})$.

Permutation tests in ANOVA. If normality cannot be assumed for the populations, or the sample is too small for the CLT approximation to an F-distribution, the null distribution of the F-statistic can be generated by shuffles. But here the permutation distribution is obtained not by shuffling the treatments of all observations, but by shuffling the treatments within each block. The reason is seen from the model when the null $H_0 : t_1 = \cdots = t_k = 0$ is true:

$$X_{ij} = \mu + b_j + \epsilon_{ij}. \tag{6.14}$$

If the ϵ_{ij} are independent with the same distribution, two observations in the first block, $X_{11} = \mu + b_1 + \epsilon_{11}$ and $X_{21} = \mu + b_1 + \epsilon_{21}$, are independent with the same distribution, hence exchangeable. But two observations in different blocks, X_{11} and X_{12}, have different distributions and are not exchangeable. The observations $X_{1j}, \ldots, X_{kj}$ within the jth block are exchangeable, but observations across blocks are not. So the observations within a block can be shuffled in $k!$ different ways, and doing this in all b blocks gives $(k!)^b$ possible shuffles.

The permutation F-test p-value is obtained as follows:

1. Compute F_{obs} from the observations.
2. Shuffle the levels of the treatment factor within each block, and compute the F-statistic f_i from the shuffled data.
3. Repeat step 2 for all $M = (k!)^b$ possible shuffles, or $M = 1999$ random shuffles, obtaining $f_1, \ldots, f_M$.
4. The permutation F-test p-value is $p = \frac{1+\sum_{i=1}^{M} I(f_i \geq F_{\text{obs}})}{1+M}$.

Example 6.12

For the (non-replicated) smart-farm data of Example 6.10, the observed F-statistic is $F_{\text{obs}} = 3.95$. The book compares the normal-theory null distribution $F_{2,4}$ with the exact permutation distribution computed from all $(k!)^b = (3!)^3 = 216$ shuffles (histogram: permutation distribution; curve: density of $F_{2,4}$). Under normality the p-value is $P(F_{2,4} \geq F_{\text{obs}}) = 0.113$, and the permutation F-test p-value is 0.139: both tests conclude there is not enough evidence that the three cultivation methods differ. The two functions `F.stat.CBD()` and `F.test2.exact()` in `np_supp.R` were used.

```
> farm = read.csv("data/smartfarm1.csv")
> # classical F-test
> (Fobs = F.stat.CBD(x, treatment, block, data = farm))
[1] 3.952191
> k = 3;b = 3
> (pval = 1-pf(Fobs,k-1, (k-1)*(b-1)))
[1] 0.1129032
>
> # exact permutation using F statistic (for b = 3 only)
> Fpermtest = F.test2.exact(x,treatment,block, data = farm)
> Fpermtest$pvalue
[1] 0.1388889
```

The permutation test with the F-statistic is exactly the same as the permutation test with the treatment variability $SS_{\text{Treatment}}$ as statistic.[5] This can be understood for the same reason that

[5] See Exercise 6.4 of the book.

comparing two populations, the permutation tests with the two-sample t-statistic and with the difference of means were the same, and comparing several populations, the permutation tests with F and with $SS_{\text{Treatment}}$ were the same. (Within each block the shuffles preserve the treatment-free total and block sums, so SS_{Total} and the block sum of squares are fixed and F is increasing in $SS_{\text{Treatment}}$.)

6.2.2 The Friedman test

The permutation test in a complete block design using as statistic the F-statistic computed after rank-transforming the observations, instead of the X_{ij}, is the distribution-free test proposed by Friedman (Friedman, 1937). Even under the null $H_0 : t_1 = \cdots = t_k$ the levels of the observations may differ by block, so the rank transformation is done separately for the k observations within each block: in the jth block, $X_{1j}, \ldots, X_{kj}$ are transformed to ranks $1, \ldots, k$.

Example 6.13

The rank transformation of the smart-farm data of Example 6.10; the ranks are assigned within blocks.

```
> farm = read.csv("data/smartfarm1.csv")
> as.data.frame(farm %>% arrange(block) %>% group_by(block) %>%
    mutate(ranks = rank(x)) )
   x treatment block ranks
1 105         A    in     2
2  96         B    in     1
3 112         C    in     3
4  85         A  over     1
5  95         B  over     2
6 100         C  over     3
7  95         A under     1
8 110         B under     2
9 117         C under     3
```

As the Kruskal–Wallis statistic did for comparing several populations without blocks, Friedman's statistic is the treatment variability of the (rank-transformed) ranks standardized by the sample variance of the ranks:

$$FM = \frac{12}{k(k+1)} b \sum_{i=1}^{k} (\bar{R}_{i\cdot} - \bar{R})^2.$$

The $\frac{k(k+1)}{12}$ in the denominator of Friedman's statistic is the sample variance of the ranks $1, \ldots, k$. With tied observations and mid-ranks, the denominator of Friedman's statistic is replaced by the average $\frac{1}{b}\sum_{j=1}^{b} S_j^2$ of the sample variances S_j^2 of the ranks in each block, and

$$FM = \frac{1}{\frac{1}{b}\sum_{j=1}^{b} S_j^2} \, b \sum_{i=1}^{k} (\bar{R}_{i\cdot} - \bar{R})^2$$

is Friedman's statistic. As for the Kruskal–Wallis statistic, it can be seen as standardized to use that for many observations, under the null,

$$FM \dot{\sim} \chi^2_{k-1}.$$

But in a complete block design with a fixed number of treatments k, the number of observations grows only as the number of blocks b grows, so in most cases where k and b are not large the large-sample null distribution is not accurate, and it is recommended to use the exact permutation distribution or a random permutation distribution as the null distribution.

Example 6.14

Testing the cultivation methods for the smart-farm data of Example 6.10, Friedman's statistic is $FM_{\text{obs}} = 4.67$, and compared with the large-sample χ^2_{3-1} the (approximate) p-value is 0.097. The function `FM.stat.CBD()` in `np_supp.R` computes Friedman's statistic.

```
> (FMobs = FM.stat.CBD(x,treatment,block,data = farm))
[1] 4.666667
> # pvalue (asymptotic)
> (pval1 = 1-pchisq(FMobs,k-1))
[1] 0.09697197
```

The large-sample p-value can also be computed with R's built-in `friedman.test()`:

```
> friedman.test(x ~ treatment | block,data = farm)

        Friedman rank sum test

data:  x and treatment and block
Friedman chi-squared = 4.6667, df = 2, p-value = 0.09697
```

The function `Friedman.test.exact()` in `np_supp.R` performs the exact permutation test:

```
> # pvalue (exact permutation test)
> Friedman.test.exact(x,treatment,block, data = farm)$pvalue
[1] 0.1944444
```

Compared with the exact null distribution, the permutation distribution, the observed statistic: of the $(3!)^3 = 216$ shuffles, 42 give statistics at least FM_{obs}, so the p-value is $42/216 = 0.194$, quite different from the large-sample p-value. This difference is because the permutation distribution of Friedman's statistic computed from ranks is very discrete. The quantile plot comparing the exact null distribution (permutation distribution) with χ^2_2 (in the book) shows that for these data with $n = 9$ the large-sample approximation of the null distribution is very inaccurate.

When considering all shuffles is hard, a random permutation test is used, generating the null distribution exactly; remember that the rank transformation and the shuffles are within blocks. The function `Friedman.test.randperm()` in `np_supp.R` implements the random permutation test with Friedman's statistic. Since the null distribution of the statistic is very discrete, the random shuffles converge slowly to the exact p-value.

```
> set.seed(0)
> Friedman.test.randperm(x, treatment, block, M = 9999, data = farm)$
    pvalue
[1] 0.1107
```

6.2.3 Tests of ordered alternatives

In the complete block model

$$X_{ij} = \mu + t_i + b_j + \epsilon_{ij}, \qquad i = 1, \ldots, k, \quad j = 1, \ldots, b, \tag{6.15}$$

if the treatments are expected to have effects in some order, an ordered alternative can be set: against the null $H_0 : t_1 = \cdots = t_k$,

$$H_1 : t_1 \le t_2 \le \cdots \le t_k \tag{6.16}$$

(with at least one inequality "$<$").

Without a block variable, the Jonckheere–Terpstra test tests hypotheses about the order of several populations, and was more powerful than the Kruskal–Wallis test (against general alternatives). Recall that the Jonckheere–Terpstra statistic measures nonparametrically the correlation between the observations X_{ij} and the population order $\{1, \ldots, k\}$.

With a block variable, the block effects are added to the observations X_{ij}. So one can compute within each block a statistic representing the correlation between the observations X_{ij} and the population order $\{1, \ldots, k\}$, and then simply add them up. If the ranks of the observations in block j are $R_{1j}, \ldots, R_{kj}$, then if the order (6.16) is true, $R_{1j} < \cdots < R_{kj}$ tends to hold, and $\sum_{i=1}^{k} iR_{ij}$ (related to Spearman's rank correlation) tends to be large. Page's (Page, 1963) L-statistic is the sum of these block statistics,

$$L = \sum_{j=1}^{b} \Big[\sum_{i=1}^{k} iR_{ij} \Big] = \sum_{i=1}^{k} iT(R_{i\cdot}), \tag{6.17}$$

where

$$T(R_{i\cdot}) = \sum_{j=1}^{b} R_{ij}$$

is the sum of the ranks of treatment i. Page proposed tests using as null distribution of this statistic the permutation distribution (generated by within-block shuffles) or its normal large-sample approximation, called **Page's trend test**. In complete block designs samples are usually small and the large-sample approximation[6] is very inaccurate, so a permutation test with L is recommended.

Example 6.15

The table below summarizes hypothetical data comparing four treatments in six blocks; the within-block ranks are in parentheses. Use Page's test to see whether the treatment effect is smallest for treatment 1 and largest for treatment 4.

	block 1	block 2	block 3	block 4	block 5	block 6	rank sum
treatment 1	128 (3)	128 (1)	160 (1)	142 (1)	157 (2)	179 (3)	11
treatment 2	122 (2)	137 (2)	177 (2)	177 (3)	160 (3)	138 (1)	13
treatment 3	207 (4)	188 (3)	181 (3)	164 (2)	155 (1)	175 (2)	15
treatment 4	120 (1)	208 (4)	199 (4)	194 (4)	177 (4)	195 (4)	21

For these data Page's L-statistic is $L = 1 \times 11 + 2 \times 13 + 3 \times 15 + 4 \times 21 = 166$, and the random permutation p-value is 0.018: the differences between treatment effects are significant. A Friedman test on these data gives $p = 0.113$, showing the lower power of Friedman's test. The function `Page.L.stat()` in `np_supp.R` computes L, and the random permutation test with this statistic is implemented in `Page.trend.randperm()`.

```
> dat = read.csv("data/toydat_page.csv")
> (Lobs = Page.L.stat(x,treatment,block, data =dat))
[1] 166
> set.seed(0)
> Page.trend.randperm(x,treatment,block, data = dat)$pvalue
[1] 0.018
```

The levels of the treatment must be ordered, so the categorical variable `treatment` must have levels in the numeric form $1, 2, \ldots, k$.

[6] For the large-sample null distribution of Page's L-statistic see Exercise 6.6 of the book. (Supplied: under H_0 the within-block rank vectors are independent uniform permutations, so $E(L) = \frac{bk(k+1)^2}{4}$ and $\mathrm{Var}(L) = \frac{bk^2(k+1)(k^2-1)}{144}$, and L is asymptotically normal as $b \to \infty$.)

6.3 General two-way layouts

Non-replicated complete block designs like Example 6.10 and replicated ones like Example 6.11 are special cases of the general two-way layout. Not only these two examples: any design without observations in every treatment-block cell, i.e. an incomplete block design, also with two factors (treatment and block, or two treatments), is called a two-way layout.

In a general two-way layout one sets up the model (6.11) with interaction, or, assuming no interaction effects, the simpler model

$$X_{ijl} = \mu + t_i + b_j + \epsilon_{ijl}, \qquad i = 1, \ldots, k, \quad j = 1, \ldots, b, \quad l = 1, \ldots, c. \tag{6.18}$$

Here the number of observations (replications) in each treatment-block cell is assumed to be c.

Without interaction as in (6.18), we compare the null $H_0 : t_1 = \cdots = t_k$ with the alternative H_1: "$t_i \neq t_j$ for some $i \neq j$." Assuming ϵ_{ijl} in (6.18) approximately normal, the F-test is the general two-way ANOVA method. If normality is hard to assume, a permutation test with the F-statistic is a distribution-free test. Then the kc observations within each block are exchangeable under the null, so, as without replication, the permutation distribution is generated by shuffling the observations within each block. (The number of all shuffles is $[(kc)!]^b$.)

As in the Friedman test, one can consider the permutation test using as statistic the F-statistic computed from the ranks of the observations within each block. The test considering rank transformation was first proposed by Skillings and Mack (Skillings and Mack, 1981) and is called the Skillings–Mack test. The Skillings–Mack statistic modifies the F-statistic slightly. Let R_{ijl} be the within-block ranks; the jth block has kc observations, so $R_{ijl} \in \{1, \ldots, kc-1, kc\}$. Dividing the rank-transformed treatment variability by the number of replications c,

$$SS_{\text{Treatment}}/c = b \sum_{i=1}^{k} (\bar{R}_{i\cdot\cdot} - \bar{R})^2,$$

the Skillings–Mack statistic standardizes it by the sample variance of the ranks:

$$SM = \frac{12}{kc(kc+1)} b \sum_{i=1}^{k} (\bar{R}_{i\cdot\cdot} - \bar{R})^2,$$

where $kc(kc+1)/12$ is the sample variance of the ranks $\{1, \ldots, kc-1, kc\}$. With ties, replacing it by the average $\frac{1}{b}\sum_{j=1}^{b} S_j^2$ of the within-block sample variances S_j^2 of the ranks,

$$SM = \frac{1}{\frac{1}{b}\sum_{j=1}^{b} S_j^2} \, b \sum_{i=1}^{k} (\bar{R}_{i\cdot\cdot} - \bar{R})^2$$

is the statistic. As for Friedman's statistic, when the null $H_0 : t_1 = \cdots = t_k$ is true, asymptotically $SM \sim \chi^2_{k-1}$, so the asymptotic p-value is $p = P(\chi^2_{k-1} \geq SM_{\text{obs}})$.

Example 6.16

With the replicated smart-farm data of Example 6.11, test whether production differs by cultivation method. The data have $k = 3$ factor levels, $b = 3$ blocks and $c = 3$ replications. The function `SM.stat()` in `np_supp.R` computes the Skillings–Mack statistic, and the random permutation test with it is implemented in `SM.perm.test()`.

```
> (SMobs = SM.stat(x,treatment,block,data = smartfarm))
[1] 4.489013
```

The test with the chi-squared large-sample approximation gives $p > 0.1$: the differences by cultivation method are not significant.

```
> (pvalue.asymp = 1 - pchisq(SMobs, df = 2))
[1] 0.1059798
```

With $n = 27$ observations, can the large-sample null distribution be trusted? The random permutation test gives a different result:

```
> SM.perm.test(x,treatment,block,M=9999,data = smartfarm)$pvalue
[1] 3e-04
```

Only 3 of the 10000 shuffles gave statistics at least the observed one, so the random permutation p-value is $p = 3/10000$: the difference by cultivation method is significant. In the code above the number of random shuffles $M = 9999$ was made rather large. Since the two tests conflict, the accuracy of the random permutation p-value needed to be increased, so the number of repetitions was made large.

In complete block designs where the number of replications per treatment-block cell is not constant, or in incomplete block designs, tests of treatment effects were developed in ways similar to the permutation tests, Friedman test and Skillings–Mack test above; see Chapter 7 of Hollander et al. (2013) for details.

Finally, when there is interaction, note that for tests of treatments or of interaction, rank transformation may remove the evidence of interaction present in the data. The interaction is a quadratic term, the product of treatment and block effects, and the nonlinear rank transformation may remove it. To make nonparametric tests in model (6.11) with interaction, it is recommended to use special rank transformations,[7] or not to rank-transform and instead regard the two-way model as a kind of linear regression model and test by resampling. (Resampling inference in linear regression is treated in Chapter 10.)

[7] The dangers of rank transformation with interaction and the special rank transformations mentioned in the text are described in detail in Section 9.2 of Higgins (2004).